

Over the past decade, sodium-glucose cotransporter-2 (SGLT2) inhibitors and glucagon-like peptide-1 (GLP-1) receptor agonists have expanded the goals of T2DM treatment beyond glycemic control. Both classes improve metabolic parameters and have demonstrated cardiovascular and kidney benefits, but their mechanisms and strongest clinical effects differ. SGLT2 inhibitors reduce renal glucose reabsorption in the proximal tubule, resulting in increased urinary glucose excretion. GLP-1 receptor agonists enhance glucose-dependent insulin secretion, suppress glucagon secretion, delay gastric emptying, and promote weight loss [4-5]. These differences provide a biological basis for their distinct and potentially complementary clinical effects.

Landmark cardiovascular outcome trials established the cardiorenal effects of both drug classes. EMPA-REG OUTCOME demonstrated reductions in cardiovascular mortality and hospitalization for heart failure with empagliflozin, while Canagliflozin Cardiovascular Assessment Study (CANVAS) and DECLARE-TIMI 58 provided additional evidence for cardiovascular, heart failure, and kidney benefits with SGLT2 inhibition [6-8]. Subsequent trials expanded these findings beyond traditional cardiovascular outcome populations. DAPA-HF and EMPEROR-Reduced demonstrated benefits in patients with heart failure with reduced ejection fraction [9-10], while DELIVER extended evidence for dapagliflozin to patients with mildly reduced or preserved ejection fraction [11]. Dedicated kidney outcome trials, including CREDENCE, DAPA-CKD, and EMPA-KIDNEY, further established the role of SGLT2 inhibitors in slowing CKD progression and reducing cardiorenal events [12-14].

GLP-1 receptor agonists have demonstrated a somewhat different pattern of benefit. LEADER and SUSTAIN-6 showed reductions in major adverse cardiovascular events (MACE) with liraglutide and semaglutide, respectively [15-16]. In addition to their cardiovascular effects, GLP-1 receptor agonists provide clinically important reductions in body weight and have increasingly demonstrated kidney benefits. The FLOW trial provided dedicated kidney outcome evidence for semaglutide in patients with T2DM and CKD, further expanding the role of this class in cardiorenal risk reduction [17]. As this evidence accumulated, treatment recommendations increasingly incorporated cardiovascular, heart failure, kidney, and metabolic risk into the selection of glucose-lowering therapy rather than relying primarily on glycemic targets [18-21]. More recent randomized and observational evidence has also examined GLP-1 receptor agonists and SGLT2 inhibitors in combination, raising the possibility that their differing mechanisms and outcome profiles may provide complementary benefits [22-28].

Despite the growing evidence supporting both classes of medications, important questions remain regarding their relative and complementary roles in patients with different cardiovascular, heart failure, kidney, and metabolic risk profiles. Much of the randomized evidence was generated in placebo-controlled cardiovascular, heart failure, or kidney outcome trials rather than direct comparisons between SGLT2 inhibitors and GLP-1 receptor agonists. Comparative observational studies and meta-analyses provide additional evidence, but differences in study populations, outcome definitions, follow-up periods, and study designs complicate interpretation. Evidence regarding combination therapy is also evolving, particularly for cardiovascular and kidney outcomes beyond those established for either class alone.

This systematic review addresses these gaps by integrating randomized controlled trials, comparative observational studies, and systematic reviews and meta-analyses within a single clinically focused framework. It incorporates recent dedicated cardiovascular, heart failure, and kidney outcome evidence while also evaluating metabolic effects, safety, real-world comparative effectiveness, and the emerging evidence for complementary use of both classes. By considering study design and risk of bias when interpreting differences across the evidence base, this review aims to clarify where the benefits of SGLT2 inhibitors and GLP-1 receptor agonists overlap, where they differ, and how these differences may inform individualized treatment decisions in adults with T2DM.

## Review

### Methods

#### *Review Design and Registration*

This systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Appendix 1). The review evaluated cardiovascular, heart failure, kidney, mortality, metabolic, and safety outcomes associated with SGLT2 inhibitors and GLP-1 receptor agonists in adults with T2DM, including evidence regarding their complementary use. Because substantial clinical and methodological heterogeneity was expected across study populations, interventions, comparators, outcome definitions, follow-up periods, and study designs, a structured narrative synthesis was used rather than a de novo meta-analysis.

The review was retrospectively registered with the Open Science Framework (OSF) on August 30, 2026 (registration DOI: 10.17605/OSF.IO/PRNYA). Earlier narrative-review work preceded registration. The subsequent systematic-review conversion included an updated literature search, deduplication, independent screening, documentation of full-text exclusion reasons, structured data extraction, risk-of-bias assessment, and PRISMA 2020 reporting.